



Dual-masked contrastive learning based hypergraph foundation model for whole slide images

Xueying Zhou^{a,1} , Saisai Ding^{a,1}, Wenhua Zhang^b, Juncheng Li^a, Jun Wang^a, Jiasheng Chen^c, Jun Shi^{a,*}

^a The Key Laboratory of Specialty Fiber Optics and Optical Access Networks, Joint International Research Laboratory of Specialty Fiber Optics and Advanced Communication, School of Communication and Information Engineering, Shanghai University, Shanghai, 200444, PR China

^b School of Future Technology, Shanghai University, Shanghai, 200444, PR China

^c Department of Urology, Shanghai Sixth People's Hospital Affiliated to Shanghai Jiao Tong University School of Medicine, Shanghai, 200233, PR China

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ABSTRACT

The spatial and contextual information in whole slide images (WSIs) is crucial for improving the diagnostic accuracy of computer-aided diagnosis (CAD). Although existing hypergraph-based methods have shown their effectiveness in capturing complex contextual dependencies among patches, they still lack the generalizability across cancer types and diagnostic tasks. Foundation models can address this issue by utilizing the strategy of self-supervised pre-training on large-scale datasets, which allows knowledge transfer across different tissue types and diseases. However, existing hypergraph self-supervised approaches have yet to effectively integrate fine-grained local tissue features with global inter-region dependencies. To this end, we propose a Dual-Masked Contrastive Learning based Hypergraph Foundation Model for WSIs. For pre-training the HFM, a novel node-hyperedge dual-masked hypergraph modeling strategy is developed. It masks parts of the node and hyperedge features to generate two augmented hypergraphs, respectively, and then trains an HFM to reconstruct the masked features for capturing local relationships among nodes and hyperedges. To enhance the model's ability to learn global dependencies in WSIs, the contrastive constraints are imposed on the augmented hypergraphs. Moreover, a max-mask strategy mitigates overfitting by preventing over-reliance on dominant features. After pre-training, the proposed DMCL-HFM can be fine-tuned for various downstream tasks, such as cancer survival prediction and subtype classification. Extensive experiments on WSI datasets from different tissues and diseases validate the effectiveness of DMCL-HFM, which consistently outperforms existing graph- and hypergraph-based approaches, as well as hypergraph pre-training approaches.

1. Introduction

Whole slide images (WSIs) are considered as the gold standard for diagnosing cancers. The deep learning-based computer-aided diagnosis (CAD) for WSIs can help pathologists improve diagnostic accuracy, consistency, and reproducibility, which has attracted considerable attention in computational pathology [1]. However, the huge size of WSIs makes it challenging for conventional deep learning models to process them directly due to high computational and memory costs [2–4]. To address the excessively high resolution in WSIs, multi-instance learning (MIL) has been introduced and extensively studied for the CAD of WSIs [5,6]. Specifically, given a WSI, the MIL-based methods first

divide it into smaller patches for feature extraction, and then aggregate these features to generate a WSI-level feature [7]. Since the spatial organization of cells and tissues is essential for identifying cancer invasion patterns, assessing tumor characteristics, and determining the disease stage [8], it is crucial for the CAD models to effectively capture the spatial and contextual relationships between patches within WSIs [3,4]. However, the MIL-based approaches generally only treat each patch independently and overlook the relationships among patches.

The graph-based methods can address this issue by applying graph convolutional networks (GCNs) to model patch relationships [9–11]. They have been widely studied in the CAD for WSIs, which effectively model pairwise relationships among image patches and achieve

* Corresponding author.

E-mail addresses: junshi@shu.edu.cn, junshi@ustc.edu (J. Shi).

¹ These authors contributed equally to this work.

promising performance in various diagnostic tasks. However, these pairwise relationships alone are insufficient, as pathological diagnosis also relies on global tissue patterns and long-range interactions [12]. It is known that hypergraphs extend the traditional graphs by connecting multiple nodes within a single hyperedge, which can model more complex and higher-order relationships. The hypergraphs have been explored in WSI analysis to capture the spatial organization of tissue structures and report improved model performance [13,14]. However, current hypergraph-based methods are typically trained in a supervised manner with the WSI-level annotations [14–16], which limits their generality across different tissue types and diagnostic tasks. Moreover, the scarcity of annotated data further hinders their broader applications in computational pathology.

Foundation models can address this issue by leveraging self-supervised pre-training on large-scale, unlabeled datasets across diverse tissues and diseases, and then quickly adapting to various downstream tasks by fine-tuning with limited training samples [17]. Various foundation models have been proposed for different medical modalities and tasks, such as USFM [18] for ultrasound image analysis, BME-X [19] for magnetic resonance imaging (MRI) and GMAI for generalist medical image analysis [17]. For WSIs, some studies have constructed Vision Transformer-based foundation models to effectively extract patch embeddings from image patches of WSIs [20,21]. These models excel at learning powerful visual representations through self-supervised learning and have shown strong transferability across datasets. However, they primarily focus on patch-level feature encoding and overlook the complex contextual and structural relationships among patches within a WSI. To the best of our knowledge, there is no hypergraph foundation model (HFM) specifically designed for WSI analysis. Therefore, it is necessary to develop a hypergraph model, as it could enhance the hypergraph's ability to generalize across different tissue types and diagnostic problems, making it more flexible and applicable to a wider range of computational pathology tasks.

Self-supervised learning (SSL) plays a critical role in developing foundation models, which allows models to learn meaningful representations from unlabeled data by leveraging the inherent structure and patterns in the data itself [22,23]. The SSL techniques have been extensively explored in hypergraph pre-training for tasks like citation classification [24] and session-based recommendation [25,26], which achieve notable success in these specific fields. These methods primarily adopt the contrastive learning for global feature alignment. As a result, they pay limited attention to the rich semantic interactions within local neighborhoods. This limitation may hinder the pre-training of HFMs in WSI analysis, since they need to capture both global and local relationships to effectively represent the complex tissue structures and spatial dependencies inherent in WSIs [27].

In this work, we propose a novel Dual-Masked Contrastive Learning based Hypergraph Foundation Model (DMCL-HFM) for WSIs. Specifically, some node and hyperedge features are randomly masked separately to generate two augmented hypergraphs, and the HFM is trained to reconstruct them to the original hypergraph to capture complex associations within WSIs. Moreover, in our contrastive learning framework, two augmented hypergraphs generated from the same original hypergraph are treated as a positive pair, while augmented hypergraphs derived from different original hypergraphs are treated as negative pairs. Furthermore, the maximum values of node and hyperedge features are masked during pre-training of the HFM to prevent overfitting. Extensive experiments on cancer survival prediction and subtype classification datasets show that the DMCL-HFM outperforms existing graph- and hypergraph-based methods, as well as hypergraph pre-training methods for WSI analysis.

The main contributions are summarized as follows.

- 1) A novel hypergraph foundation model, DMCL-HFM, is developed for WSI analysis, which is pre-trained on unlabeled WSIs from a wide spectrum of cancer types. It captures both local tissue patterns and

global region relationships, and can be transferred to various downstream tasks like subtype classification and survival prediction.

- 2) A node-hyperedge dual-masked contrastive learning framework is introduced to pre-train the HFM, which enables joint modeling of local patch-level semantics and global inter-region dependencies. By independently masking and contrasting node and hyperedge features, the method enhances the model's ability to learn rich structural representations from unlabeled WSIs across diverse disease types.

2. Related work

2.1. Graph-based WSI representation learning

Graph modeling has been widely used for WSI analysis, as it can capture the complex correlations among patches [28–30]. For example, Chen et al. [9] utilized patches as nodes and constructed edges by connecting adjacent patches for survival prediction. Hou et al. [31] proposed a multi-scale heterogeneous graph to learn the hierarchical representations from WSIs. Chan et al. [11] also proposed a heterogeneous graph to capture the relationships among different types of nuclei for WSI analysis. Recently, several works have extended the graph to hypergraph structure to process WSIs with highly complicated data correlation [13,14,32,33]. Di et al. [14] proposed a multi-hypergraph-based MIL framework to generate the high-order representation via multilateral correlation modeling within WSIs. Cai et al. [33] present a multimodal hypergraph neural network that integrates both morphological and genetic information for survival analysis. These graph- and hypergraph-based methods have proven effective in modeling complex spatial and semantic relationships in WSIs, which demonstrates the potential of hypergraph modeling to capture rich contextual and structural information.

Although the hypergraphs effectively capture complex relationships within intricate WSIs, they heavily rely on annotated data. For rare cancers, this dependency can hinder model performance due to the limited availability of samples.

2.2. Self-Supervised learning for medical foundation models

Foundation models are trained on massive and diverse medical datasets, which allows them to be applied to a wide range of downstream tasks [17,34]. These models undergo self-supervised pre-training, where they learn rich, transferable representations without relying on task-specific annotations [22,23,35]. The SSL paradigm helps models capture underlying patterns and structures in the data through pretext tasks. Once pre-trained, these models can be fine-tuned for specific tasks, making them highly effective for various applications, particularly in scenarios where annotated data are limited or costly to obtain, such as medical image analysis.

In recent years, several medical foundation models have been developed, leveraging SSL techniques to enhance performance across various healthcare applications [18,21,36]. The Med-BERT [36] uses Masked Language Modeling (MLM) for self-supervised pre-training on clinical text, capturing rich representations of medical terminology. It is applied to tasks such as disease prediction, clinical decision support, and patient outcome forecasting. USFM [18] proposes a novel MIM-based spatial-frequency dual masking method in the self-supervised pre-training for ultrasound foundation models, which tackles the challenges of low quality and feature ineffectiveness. Prov-GigaPath [21] proposes a novel vision transformer architecture for pretraining gigapixel pathology slides. Its pretraining begins with patch-level self-supervised learning using the classical contrastive learning method DINOv2 [37] and a standard Vision Transformer (ViT). This is followed by whole-slide-level self-supervised learning using a masked autoencoder (MAE), enabling the model to capture both local and global representations effectively. Despite the widespread application of self-supervised

foundation models, no research, to the best of our knowledge, has been conducted on hypergraph-based foundation models for WSIs.

3. Methodology

Given M unlabeled WSI datasets $\mathcal{D} = \{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_M\}$, our proposed DMCL-HFM aims to pre-train a hypergraph encoder network as an HFM φ_F through our proposed dual-masked contrastive learning. By comprehensively capturing the global contextual information of WSIs and the complex relationships among patches, the pre-trained HFM φ_F can be fine-tuned for a wide range of downstream tasks, including cancer subtype classification and survival prediction.

As illustrated in Fig. 1, the proposed DMCL-HFM comprises the following main components:

- 1) WSI Pre-Processing & Hypergraph Construction: Each WSI is first divided into numerous non-overlapping patches, which are then used to construct a hypergraph based on the Euclidean distances of the patch embeddings.
- 2) Dual-Masked Hypergraph Contrastive Learning: For pre-training the HFM, the hypergraph's node and hyperedge features are separately masked at random to create two augmented hypergraphs. These augmented hypergraphs are then processed through an encoder and decoder in sequence to reconstruct the masked features. Additionally, augmented hypergraphs from the same hypergraph are positive pairs, while those from different hypergraphs are negative pairs. The similarity of positive pairs is maximized, while that of negative pairs is minimized. To prevent overfitting, the highest-valued node and hyperedge features are masked during the learning process.
- 3) Fine-Tuning for Downstream Tasks: The pre-trained HFM can be transferred to various downstream tasks, including WSI classification and regression tasks such as cancer subtype classification and survival prediction.

3.1. WSI pre-processing & hypergraph construction

To address the huge size of WSIs, we adopt the conventional pre-processing workflow, which mainly consists of three procedures: tissue segmentation, patch cropping, and feature extraction. Given a WSI, to eliminate the interference from background regions, we first use the Otsu algorithm to segment the tissue regions from it. Subsequently, these segmented tissues are cropped into N non-overlapping 299×299 patches, obtaining a patch set $I_p = \{i_1, i_2, \dots, i_N\}$, where $i \in \mathbb{R}^{299 \times 299 \times 3}$. Lastly, a pre-trained CTransPath [38] is employed to convert each patch i into a feature vector v with 768 dimensions. To obtain the node representations of WSI, we aggregate N patch features and form a node set $\mathcal{V} = \{v_1, v_2, \dots, v_N\} \in \mathbb{R}^{N \times 768}$.

The graphs and hypergraphs are effective tools for modeling data relationships. The traditional graphs use edges to capture pairwise

relationships, which makes them simple and efficient. However, WSIs involve multi-region spatial and semantic interactions, which limit traditional graphs to only two-node connections. The hypergraphs address this limitation by using hyperedges that can connect multiple nodes simultaneously. This allows them to model higher-order relationships, such as capturing group interactions among tissue patches, which reveal complex dependencies crucial for understanding tissue organization and disease progression.

A hypergraph is composed of a set of nodes and a collection of hyperedges [13], and can be defined as $\mathcal{H} = (\mathcal{V}, \mathcal{E}, W)$. Here, $\mathcal{V} \in \mathbb{R}^{N \times L}$ is the node set, $\mathcal{E} \in \mathbb{R}^{E \times L}$ is the hyperedge set, and $W \in \mathbb{R}^E$ is the hyperedge weight matrix, where N and E denote the number of nodes and hyperedges, respectively, while L represents the dimension of node and hyperedge features. To describe the complex correlations in a hypergraph, an incidence matrix $H \in \mathbb{R}^{N \times E}$ is used to represent the connection relationships between nodes, which can be denoted as:

$$H(v, e) = \begin{cases} 1, & \text{if } v \in e \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

After the processing of WSIs, the hyperedge construction becomes an essential step in utilizing the hypergraph to learn the complex relationships among patches. To enhance the representation of intricate connections between patches, we utilize the spatial coordinates of patches to construct E hyperedges. Specifically, each patch is connected to its k -nearest patches based on their spatial positions within a WSI, which enables the capture of local spatial information. After that, we obtain the incidence matrix $H \in \mathbb{R}^{N \times E}$, which is used to generate hyperedge set $\mathcal{E} = \{e_1, e_2, \dots, e_E\}$ ($e \in \mathbb{R}^{768}$ following HGNN+ [13]). Finally, a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ with the incidence matrix H is constructed for each WSI as the input of the HFM.

3.2. Dual-Masked hypergraph contrastive learning

The overall framework of our proposed Dual-Masked Hypergraph Contrastive Learning (DMCL) approach is illustrated in Fig. 2. It consists of two main components: (1) the node-hyperedge dual-masked hypergraph modeling, which captures local structural dependencies among patch embeddings within WSIs, and (2) the max-masking contrastive learning framework, which learns global semantic representations of WSIs while enhancing the robustness of the learned features.

3.2.1. Node-Hyperedge dual-masked hypergraph modeling

To enable the HFM to better capture the complex local relationships between different image patches in WSIs, we propose a dual-masked reconstruction mechanism. On one hand, some node features are masked and subsequently reconstructed using the hyperedge features and the incidence matrix. On the other hand, some hyperedge features are masked and reconstructed by leveraging the node features and the incidence matrix.

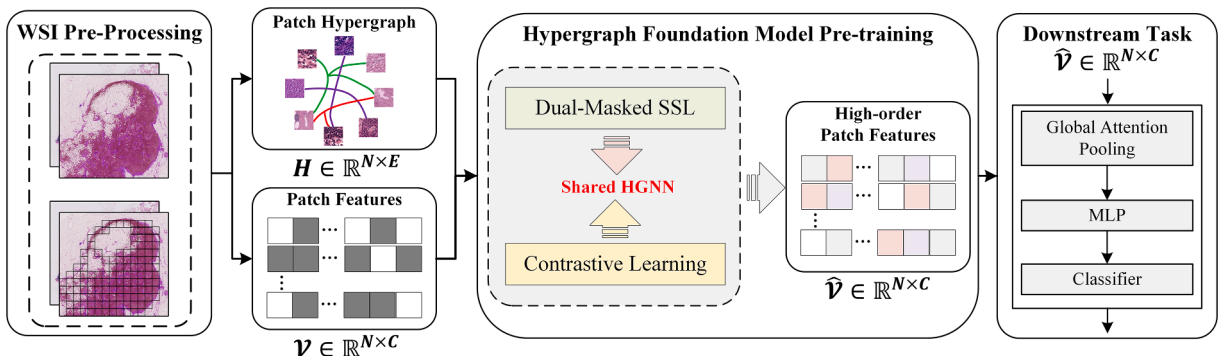


Fig. 1. Overview architecture of proposed DMCL-HFM.

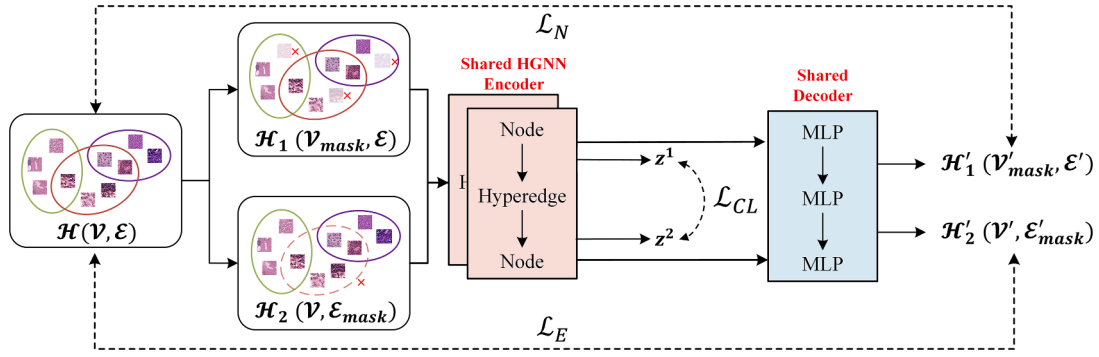


Fig. 2. Flowchart of the proposed dual-masked hypergraph contrastive learning approach.

Specifically, for a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, we separately mask parts of the nodes and hyperedges to obtain the two augmented versions of the hypergraph $\mathcal{H}_1 = (\mathcal{V}_{mask}, \mathcal{E})$ and $\mathcal{H}_2 = (\mathcal{V}, \mathcal{E}_{mask})$. If a node or hyperedge is selected to be masked, all the values of its features (i.e., v and e) are set to zeros. The masking function $Mask(\cdot)$ can be expressed as:

$$Mask(f_j) = \begin{cases} 0, & \text{if } j \in \text{masked features} \\ f_j, & \text{otherwise} \end{cases}, \quad (2)$$

where $f \in \mathbb{R}^{1 \times 768}$ is a feature vector associated with a node or hyperedge in the hypergraph. Under a fixed masking ratio of k_1 , the selection of feature vectors for masking occurs randomly.

All the augmented hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$ are then fed into a hypergraph encoder network φ_F along with the same incidence matrix for encoding, and are subsequently reconstructed by the decoder φ_D , obtaining $\mathcal{H}'_1 = (\mathcal{V}'_{mask}, \mathcal{E})$ and $\mathcal{H}'_2 = (\mathcal{V}', \mathcal{E}'_{mask})$. The network architecture of φ_F can be any hypergraph neural network, and in this work, we use HGNN+ [13] as our backbone. φ_D is composed of a multilayer perceptron (MLP), with a ReLU activation function following each linear layer.

To achieve node-hyperedge dual-masked hypergraph modeling, two reconstruction loss functions are employed: one for node reconstruction and one for hyperedge reconstruction, which are formulated as follows:

$$\mathcal{L}_N = \frac{1}{\sum_{j=1}^N m_j} \sum_{j=1}^N (m_j |v_j - v'_j|), \quad (3)$$

$$\mathcal{L}_E = \frac{1}{\sum_{j=1}^E m_j} \sum_{j=1}^E (m_j |e_j - e'_j|), \quad (4)$$

where m_j is the mask value for the j -th element (1 if included in the masked feature vectors, 0 if excluded from them). v_j and e_j are the feature vectors of $\mathcal{V}'_{mask}$ and $\mathcal{E}'_{mask}$, respectively. To make the HFM more effectively learn the intricate dependencies between image patches and enhance its ability to model the complex relationships within WSIs, the two loss functions are fused as follows:

$$\mathcal{L}_{DM} = \lambda_N \mathcal{L}_N + \lambda_E \mathcal{L}_E. \quad (5)$$

3.2.2. Max-masking contrastive learning

Although the proposed node-hyperedge dual-masked hypergraph modeling strategy enables the HFM to capture complex relationships within local neighborhoods in WSIs, it still lacks the ability to learn global relationships. To address this, we introduce a contrastive learning framework that pulls together the global features of positive samples while pushing apart those of negative samples. In particular, the two hypergraphs $\mathcal{H}_1 = (\mathcal{V}_{mask}, \mathcal{E})$ and $\mathcal{H}_2 = (\mathcal{V}, \mathcal{E}_{mask})$, obtained from the same hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, serve as two different data perturbations in contrastive learning, forming a set of positive sample pairs. In

contrast, pairs obtained from different hypergraphs are treated as negative sample pairs.

To map hypergraphs into the global contrastive learning space, all hypergraphs are first passed through the hypergraph encoder network φ_F with their incidence matrix to extract node features $f_v \in \mathbb{R}^{N \times 512}$. Subsequently, a global attention pooling mechanism [6] is applied to aggregate all node features into a global feature $g \in \mathbb{R}^{1 \times 512}$ for each hypergraph. Finally, a projection head network φ_P is constructed to obtain the contrastive learning embedding $z \in \mathbb{R}^{1 \times 128}$. Our projection head network φ_P is also implemented as MLP, where each linear layer is followed by a ReLU activation function.

Furthermore, to mitigate the issue of majority feature deactivation caused by excessive contribution of certain features to global feature extraction, thereby enabling more comprehensive extraction of global information, the top k_2 -percentage largest element values within the output node and hyperedge features at each HGNN layer of the HFM are masked, which is expressed as:

$$Maxmask(s) = \begin{cases} 0, & s \geq t_s \\ s, & s < t_s \end{cases}, \quad (6)$$

where $s \in \mathbb{R}^1$ is an element of a node or hyperedge feature, and t_s represents the value threshold for the largest k_2 -percentage of all elements in the feature.

To perform contrastive learning for modeling the global features of WSI images, a minibatch of B hypergraphs is randomly sampled from datasets and processed to obtain 2B contrastive learning embeddings. Subsequently, the NT-Xent loss [39] is used for model training and is formulated as follows:

$$\mathcal{L}_{CL} = -\log \frac{\exp(\text{sim}(z_b^1, z_b^2)/\tau)}{\sum_{b=1, b' \neq b}^B \exp(\text{sim}(z_b^1, z_{b'}^2)/\tau)}, \quad (7)$$

where $\exp(\cdot)$ denotes the exponential function, and $\text{sim}(\cdot)$ is the cosine similarity function. z_b^1, z_b^2 are the contrastive learning embeddings of the two augmented hypergraphs $\mathcal{H}_1$ and $\mathcal{H}_2$ for the b -th hypergraph $\mathcal{H}$ in the minibatch B , and τ represents the temperature parameter.

In total, to simultaneously model the global features of WSI images and the complex internal relationships between their patches, the overall loss function is defined as follows:

$$\mathcal{L}_{Total} = \mathcal{L}_{DM} + \alpha \mathcal{L}_{CL}, \quad (8)$$

3.3. Fine-tuning for downstream tasks

3.3.1. Multiple instance learning

Multiple Instance Learning (MIL) is a weakly-supervised learning framework that works on set-based data structures, where these structures are often referred to as bags. Each bag consists of a collection of instances [5]. In the case of a single-label classification task, a bag $X = \{x_1, \dots, x_m\} \in \mathbb{R}^{M \times d_m}$ consists of m instances, each represented by a

d_m -dimensional feature vector. The goal is to learn a permutation-invariant function $\mathcal{F}$ that can predict the label of the bag without instance annotations. The MIL framework can be generally expressed as:

$$\mathcal{F}(X) = \zeta(\delta(\{\theta(x_i) : x_i \in X\})). \quad (9)$$

where θ is an instance-level function that processes instance features, δ is an aggregation function that pools the instance features to a single bag-level representation, and ζ is a bag-level classifier that further processes the bag-level representation to get the predicted probability. In WSI-level prediction tasks, patches can be regarded as instances and the entire slide as a bag. The proposed HFM φ_F enhances the extraction of patch-level features, and the node features output by the HFM φ_F are used as the input to the aggregation function δ in the MIL framework.

3.3.2. Fine-Tuning hypergraph foundation models based on MIL

With the pre-trained HFM φ_F , DMCL-HFM can effectively capture the latent high-order correlations within the individual complicated WSI. Then, we fine-tune the pre-trained HFM for the downstream slide-level classification and survival prediction tasks.

The fine-tuning model consists of a hypergraph encoder, a MIL aggregation layer, and a classifier with two fully connected layers. After the hypergraph structure modeling, we input the individual node features $v \in \mathbb{R}^{N \times C}$ and incidence matrix $H \in \mathbb{R}^{N \times E}$ into the hypergraph encoder to learn the high-order information of WSIs and obtain the updated node features $\hat{v} \in \mathbb{R}^{N \times C}$. Then, to get the slide-level representation $Z \in \mathbb{R}^{1 \times C}$, we utilize attention-based MIL pooling [6] to aggregate all instance features, which can be defined as:

$$Z = \sum_{i=1}^N a_i v_i \text{ with } a_i = \frac{\exp\{W_a^T \tanh(W_v v_i)\}}{\sum_{j=1}^N \exp\{W_a^T \tanh(W_v v_j)\}}, \quad (10)$$

where W_a and W_v are the learnable parameters, $\tanh$ is the activation function, N is the number of patches extracted from a WSI. Finally, the slide-level representation Z is fed into the classifier to obtain the prediction probability. For the slide-level classification task, we adopt the cross-entropy loss:

$$\mathcal{L}_{CE} = -\frac{1}{Num} \sum_{i=1}^{Num} \sum_{c=1}^C y_i \log(p_i), \quad (11)$$

where C is the number of classes, y_i is the true label, p_i is the prediction probability for class i th, and Num is the number of patients. For the survival prediction task, we adopted the negative Cox log partial likelihood loss:

$$\mathcal{L}_{NLL} = \sum_{i=1}^{Num} \delta_i \left(-s_i^p + \log \sum_{j \in \{j: s_j^g \leq s_i^g\}} \exp(-s_j^p) \right), \quad (12)$$

where s_i^p and s_i^g denote the prediction probability and ground truth of observed time, respectively. $\delta_i \in \{0, 1\}$ is the censorship status. Num is the number of comparable pairs, derived from the number of patients.

4. Experiments

4.1. Datasets

To evaluate the effectiveness of our proposed method on slide-level subtype classification and survival prediction tasks, we utilized six public cancer datasets from The Cancer Genome Atlas (TCGA). TCGA is a comprehensive cancer research consortium that provides matched clinical records, diagnostic whole slide images (WSIs), and genomic data with annotated survival outcomes and censorship information. The datasets used in this study include:

TCGA-BRCA (Breast Invasive Carcinoma) dataset comprises two breast cancer subtypes, Invasive Ductal Carcinoma (IDC) and Invasive Lobular Carcinoma (ILC). It consists of 787 IDC slides and 168 ILC slides, totaling 955 diagnostic slides. This dataset is used for binary subtype classification, and the IDC samples are further used for survival prediction.

TCGA-NSCLC (Non-Small Cell Lung Cancer) dataset consists of two subtypes, Lung Adenocarcinoma (LUAD) and Lung Squamous Cell Carcinoma (LUSC). It contains 515 LUAD slides and 484 LUSC slides, totaling 999 slides. It is used for binary subtype classification. Both LUAD and LUSC samples are also used for survival prediction.

TCGA-RCC (Renal Cell Carcinoma) dataset comprises three renal cancer subtypes: Kidney Renal Clear Cell Carcinoma (KIRC), Kidney Renal Papillary Cell Carcinoma (KIRP), and Kidney Chromophobe (KICH). It includes 518 KIRC slides, 297 KIRP slides, and 121 KICH slides, totaling 936 diagnostic slides. This dataset is used for three-class subtype classification, and the KIRC and KIRP samples are used for survival prediction.

TCGA-ESCA (Esophageal Carcinoma) dataset consists of 153 diagnostic WSIs from patients with esophageal carcinoma, a malignant tumor that arises from the epithelial cells of the esophagus, and is used for survival prediction.

TCGA-LIHC (Liver Hepatocellular Carcinoma) dataset includes 287 diagnostic WSIs from patients with hepatocellular carcinoma (HCC), the most common type of primary liver cancer. This dataset is used for survival prediction.

TCGA-UCEC (Uterine Corpus Endometrial Carcinoma) dataset includes 387 diagnostic WSIs from patients with endometrial carcinoma, a common gynecologic malignancy that originates from the epithelial lining of the uterine corpus. This dataset is used for survival prediction.

In total, the TCGA-BRCA, TCGA-NSCLC, and TCGA-RCC datasets comprise nearly 3000 diagnostic slides and were used as internal datasets for hypergraph self-supervised pretraining to obtain a HFM. In contrast, the TCGA-ESCA, TCGA-LIHC, and TCGA-UCEC datasets were not involved in the pretraining stage and are considered external datasets, which are used to evaluate the generalization ability of the proposed model. Subsequently, we evaluate the pre-trained foundation model on subtype classification and survival prediction tasks across all 6 datasets.

For each task, we performed five-fold cross-validation. In each fold, 80 % of the data was used for training and 20 % for testing, ensuring that every sample was tested exactly once. Within the training set of each fold, 20 % of the samples were further set aside as a validation set for model selection.

4.2. Experimental settings

We first select state-of-the-art (SOTA) graph/hypergraph-based methods (G/HG-based) as comparison benchmarks to evaluate the effectiveness of our proposed DMCL-HFM. The details of these methods are as follows:

- 1) DeepGraphSurv [40]: It employs a spectral GCN to learn the topological relationships of WSIs. It takes the randomly sampled patches as nodes, and the edges are constructed using a Euclidean distance threshold.
- 2) Patch-GCN [9]: It uses a context-aware, spatially-resolved patched based GCN to model both local and global level topological structures of WSIs, and it uses the spatial coordinates of WSIs to connect the edges between the adjacent patches.
- 3) RankSurv [41]: It adopts traditional hypergraph neural network to achieve a high-order representation for survival prediction on WSI data.
- 4) HGCN [13]: It is a general hypergraph convolutional network based model. Similar to RankSurv, it takes the randomly sampled patches as nodes, and each hyperedge is constructed using the K-nearest

neighbor method with the Euclidean distance.

Furthermore, we compare different hypergraph self-supervised pre-training methods based on the same hypergraph-based baseline (HSSL-based). The details of the baseline and these pre-training methods are as follows:

- 5) Baseline (random): It is a baseline that employs a three-layer hypergraph encoder with randomly initialized weights, without using any task-specific pretraining. It serves as a reference to assess the contribution of the proposed pretraining strategy.
- 6) DHCN [25]: It is a hypergraph pre-training method that constructs a dual-branch framework combining a hypergraph network and a graph network, with contrastive learning applied between the two branches.
- 7) HCCF [26]: It is a hypergraph pre-training method which builds a dual-branch framework where the graph network randomly drops edges to capture local information, while the hypergraph network models global information, with contrastive learning aligning the two.
- 8) HyperGCL [24]: It is a hypergraph pre-training method where one branch randomly drops hyperedges, while the other generates new hypergraphs using a hypergraph variational autoencoder. Contrastive learning is then applied between the two branches.

We performed ablation experiments to evaluate the effectiveness of key components in DMCL-HFM. Specifically, we investigated the impact of the Max-Mask, contrastive loss, and reconstruction loss by removing each component. Additionally, we tested different masking strategies, including Dual-Mask, Node-Mask, and Hyperedge-Mask, to assess the contribution of each to model performance. The details of these variants are given as follows:

- 1) DMCL-HFM w/o Max-Mask: Pre-training the HFM without the max-mask strategy.
- 2) DMCL-HFM w/o Contrast. Loss: Pre-training the HFM without contrastive loss.
- 3) DMCL-HFM w/o Recons. Loss: Pre-training the HFM without reconstruction loss.
- 4) Node-Mask: Pre-training the HFM by reconstructing the randomly masked node features.
- 5) Hyperedge-Mask: Pre-training the HFM by reconstructing the randomly masked hyperedge features.
- 6) Dual-Mask: Pre-training the HFM by reconstructing the randomly masked node and hyperedge features simultaneously.

4.3. Evaluation metrics

For the binary classification tasks on BRCA and NSCLC, we used Accuracy (ACC) and Area Under the Curve (AUC) as evaluation metrics. For the three-class classification task on RCC, we used ACC and Matthews Correlation Coefficient (MCC). For survival prediction tasks, all methods were evaluated using 5-fold cross-validated concordance index (c-Index). The c-Index is a widely used metric for evaluating the predictive accuracy of survival models, where a higher value indicates better performance. In this study, a 5-fold cross-validation procedure was conducted to evaluate the performance of each method, and the mean $\pm$ std results was reported.

4.4. Implementation details

A total of nearly 3000 H&E-stained diagnostic slides from the TCGA-BRCA, TCGA-NSCLC, and TCGA-RCC datasets were first divided into patches. Then, features were extracted from these patches using a general pre-trained feature extractor. Subsequently, 5191,946 patches at $10\times$ magnification were used for self-supervised pre-training of the DMCL-HFM model with a 3-layer hypergraph encoder. The hidden dimension of the hypergraph encoder is tuned over {128, 256, 512}. For model

optimization, we set the hyperparameters as follows: $\lambda_N = \lambda_E = 0.5$, $\alpha = 0.1$. We trained these models for 400 epochs using the AdamW optimizer with a batch size of 64. The learning rate was searched over {0.01, 0.001, 0.0001}. Following pre-training, we conducted experiments on several slide-level classification and survival prediction tasks across different datasets in the TCGA. We used the pre-trained weights to initialize the hypergraph encoders, and were fine-tuned together with the downstream MIL aggregation layers. These models were fine-tuned for 50 epochs using the AdamW optimizer with a batch size of 32. The learning rate was selected from {0.0001, 0.00005, 0.00001}. All experiments were implemented by PyTorch on a platform with three NVIDIA GeForce RTX 3090 GPUs.

5. Experimental results

5.1. Results of slide-level classification

The classification results of different methods are summarized in Table 1. Overall, DMCL-HFM consistently achieved the best performance across all datasets. Specifically, DMCL-HFM achieves the highest average accuracy of 88.87 % and an AUC of 95.07 % on the TCGA-BRCA dataset. Compared to other methods, it offers improvements of at least 1.20 % and 0.81 % on these metrics, respectively. On the TCGA-NSCLC dataset, DMCL-HFM also surpasses all competing algorithms with the top accuracy of 91.51 % and an AUC of 93.37 %, achieving gains of 1.14 % and 1.23 % on the corresponding metrics. Similarly, for the TCGA-RCC dataset, DMCL-HFM attains the best average accuracy of 93.74 % and MCC of 89.37 %, exceeding other algorithms by at least 1.39 % and 1.28 % on these metrics, respectively. Furthermore, pre-training methods like DHCN and HyperGCL consistently outperformed MIL methods across most metrics on the three datasets. This demonstrates that self-supervised pre-training strategies can effectively leverage the available data in an unsupervised manner, enabling the model to better capture underlying patterns and enhance overall performance. Nevertheless, our DMCL-HFM still outperforms these pre-training methods by integrating the novel node-hyperedge dual-masked mechanism, which ensures that complex spatial dependencies and contextual interactions are fully captured. Moreover, our DMCL-HFM exhibits a lower standard deviation compared to existing hypergraph-based approaches like HGCN and HCCF, which indicates more stable and reliable results. This reduced variability is attributed to the use of multiple constraints, including reconstruction and contrastive losses, which guide the model toward more consistent representations. As a result, DMCL-HFM is less sensitive to dataset-specific fluctuations and maintains robust performance across diverse classification tasks.

5.2. Results of survival prediction

5.2.1. Survival prediction results for internal datasets

For survival prediction tasks, where capturing context-aware relationships is crucial, we observe significantly greater performance gains compared to slide-level classification tasks, as shown in Table 2. Overall, DMCL-HFM achieves the best c-Index performance across the IDC, KIRC, KIRP, LUAD, and LUSC cancer types. Among these cancer types, DMCL-HFM shows the most significant improvement in KIRP with a c-Index score of 0.602, exceeding other algorithms by at least 0.026. Similarly, in LUAD, DMCL-HFM also achieves an impressive c-Index score of 0.568. We attribute the high performance of DMCL-HFM to three aspects. Firstly, DMCL-HFM can use local message passing to better learn context-aware features of WSIs. Secondly, compared with slide-level classification tasks, survival prediction tasks have smaller sample sizes, and therefore benefit more from the pre-trained hypergraph base model. Finally, the max-mask strategy in DMCL-HFM alleviates the overfitting of the model to a certain extent, thereby further improving the performance of the model on small sample datasets. Furthermore, in the survival prediction task, DMCL-HFM achieves the lowest variance

Table 1

The classification results of the comparison methods. The best ones are in bold.

Methods		TCGA-BRCA		TCGA-NSCLC		TCGA-RCC	
		ACC	AUC	ACC	AUC	ACC	MCC
G/HG-based	DeepGraphSurv	85.07 $\pm$ 1.10	92.87 $\pm$ 2.11	88.27 $\pm$ 1.13	90.39 $\pm$ 1.21	90.37 $\pm$ 1.23	83.32 $\pm$ 1.91
	Patch-GCN	84.82 $\pm$ 1.31	93.21 $\pm$ 1.91	89.31 $\pm$ 1.32	91.21 $\pm$ 1.98	91.17 $\pm$ 1.32	84.93 $\pm$ 1.36
	RankSurv	86.54 $\pm$ 1.76	93.74 $\pm$ 1.16	89.45 $\pm$ 1.74	90.52 $\pm$ 1.14	91.89 $\pm$ 1.19	86.07 $\pm$ 1.53
	HGCN	85.63 $\pm$ 2.03	93.92 $\pm$ 1.42	89.93 $\pm$ 2.03	91.53 $\pm$ 1.43	92.18 $\pm$ 1.72	86.52 $\pm$ 1.62
HSSL-based	Baseline (random)	85.82 $\pm$ 1.72	93.52 $\pm$ 1.92	89.63 $\pm$ 1.12	90.21 $\pm$ 1.93	92.42 $\pm$ 1.54	87.02 $\pm$ 1.83
	DHCN	87.52 $\pm$ 2.10	94.17 $\pm$ 1.83	89.72 $\pm$ 1.71	91.85 $\pm$ 1.54	93.08 $\pm$ 1.18	88.38 $\pm$ 1.74
	HCCF	86.08 $\pm$ 1.63	93.32 $\pm$ 2.41	88.42 $\pm$ 0.91	90.93 $\pm$ 1.82	91.66 $\pm$ 1.23	85.63 $\pm$ 2.31
	HyperGCL	87.67 $\pm$ 1.42	94.26 $\pm$ 1.92	90.37 $\pm$ 0.89	92.14 $\pm$ 1.27	92.35 $\pm$ 1.19	88.09 $\pm$ 1.86
Ours	DMCL-HFM	88.87 $\pm$ 1.23	95.07 $\pm$ 1.70	91.51 $\pm$ 1.55	93.37 $\pm$ 1.46	93.74 $\pm$ 1.80	$\pm$ 1.01

Table 2

The CI values of the comparison methods. The best ones are in bold.

Methods		IDC	KIRC	KIRP	LUAD	LUSC
G/HG-based	DeepGraphSurv	0.671 $\pm$ 0.015	0.646 $\pm$ 0.014	0.542 $\pm$ 0.015	0.525 $\pm$ 0.013	0.605 $\pm$ 0.014
	Patch-GCN	0.679 $\pm$ 0.013	0.637 $\pm$ 0.018	0.549 $\pm$ 0.018	0.532 $\pm$ 0.019	0.607 $\pm$ 0.014
	RankSurv	0.683 $\pm$ 0.014	0.650 $\pm$ 0.014	0.558 $\pm$ 0.021	0.539 $\pm$ 0.017	0.632 $\pm$ 0.015
	HGCN	0.694 $\pm$ 0.012	0.661 $\pm$ 0.021	0.561 $\pm$ 0.020	0.549 $\pm$ 0.019	0.624 $\pm$ 0.016
HSSL-based	Baseline (random)	0.698 $\pm$ 0.014	0.656 $\pm$ 0.014	0.562 $\pm$ 0.013	0.541 $\pm$ 0.021	0.630 $\pm$ 0.018
	DHCN	0.707 $\pm$ 0.019	0.673 $\pm$ 0.023	0.566 $\pm$ 0.013	0.556 $\pm$ 0.024	0.643 $\pm$ 0.028
	HCCF	0.691 $\pm$ 0.021	0.658 $\pm$ 0.019	0.555 $\pm$ 0.014	0.532 $\pm$ 0.018	0.579 $\pm$ 0.017
	HyperGCL	0.714 $\pm$ 0.013	0.671 $\pm$ 0.017	0.576 $\pm$ 0.019	0.553 $\pm$ 0.021	0.646 $\pm$ 0.038
Ours	DMCL-HFM	0.725 $\pm$ 0.018	0.684 $\pm$ 0.013	0.602 $\pm$ 0.013	0.568 $\pm$ 0.011	0.653 $\pm$ 0.014

across nearly all datasets. These results suggest that DMCL-HFM learns stable and generalizable representations, making it more reliable for survival analysis across diverse patient cohorts.

To further assess the patient stratification capability, we employed Kaplan-Meier (KM) analysis. For each type of cancer, patients were divided into high-risk and low-risk categories based on the median score generated by our predictive models. An accurate survival prediction would result in notably distinct KM curves between these two groups. We utilized the Log-rank test to calculate the p-value, where a smaller p-value indicates superior performance. As illustrated in Fig. 3, HyperGCL demonstrates state-of-the-art (SOTA) performance across three types of cancers: IDC, KIRP, and LUSC. Meanwhile, DHCN achieves SOTA results for KIRC and LUAD, respectively. Notably, the P-values obtained from our method for all five cancer cohorts were significantly lower compared to those of the aforementioned SOTA methods, highlighting the superiority of our model in patient stratification.

5.2.2. Survival prediction results for external datasets

To evaluate the generalization ability of DMCL-HFM, we conducted additional experiments on three external cancer datasets from TCGA, which were not used during the pretraining stage. The C-Index scores of DMCL-HFM and baseline methods on these datasets are summarized in Table 3. Overall, DMCL-HFM consistently outperformed all comparison methods across the three cancer types. Notably, the improvement was most significant on the TCGA-ESCA dataset, where DMCL-HFM achieved a C-Index of 0.676, exceeding the best baseline by at least 0.015. The relatively small sample size of ESCA suggests that our DMCL-HFM benefits from pretraining under low-data scenarios. In TCGA-LIHC and TCGA-UCEC, DMCL-HFM also achieved strong results, with C-Index values of 0.688 and 0.697, respectively. These results demonstrate the robustness and generalization capability of DMCL-HFM across external datasets with different cancer types and data distributions.

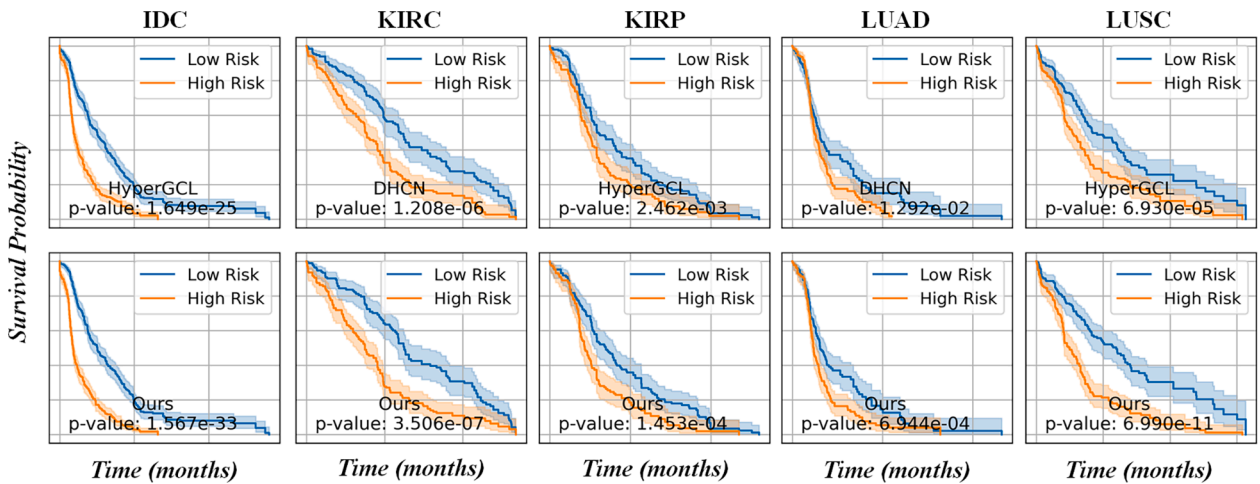


Fig. 3. KM analysis of the best comparison method and our proposed model on five datasets. For each dataset, patients from five test folds were merged for analysis. All patients were divided into high-risk (blue curves) and low-risk (red curves) groups based on the median of output risk scores. The P-values of log-rank test are shown in the bottom left.

Table 3

The survival prediction results of the comparison methods on external datasets.

Methods		LIHC	ESCA	UCEC	Average C-Index
G/HG-based	DeepGraphSurv	0.621 $\pm$ 0.015	0.607 $\pm$ 0.018	0.623 $\pm$ 0.017	0.617
	Patch-GCN	0.632 $\pm$ 0.008	0.618 $\pm$ 0.016	0.650 $\pm$ 0.018	0.633
	RankSurv	0.639 $\pm$ 0.014	0.624 $\pm$ 0.011	0.659 $\pm$ 0.014	0.641
	HGCN	0.656 $\pm$ 0.019	0.641 $\pm$ 0.015	0.669 $\pm$ 0.007	0.655
HSSL-based	Baseline (random)	0.661 $\pm$ 0.017	0.640 $\pm$ 0.010	0.670 $\pm$ 0.010	0.657
	DHCN	0.673 $\pm$ 0.009	0.652 $\pm$ 0.018	0.681 $\pm$ 0.019	0.669
	HCCF	0.667 $\pm$ 0.018	0.647 $\pm$ 0.013	0.677 $\pm$ 0.012	0.664
	HyperGCL	0.676 $\pm$ 0.017	0.661 $\pm$ 0.017	0.689 $\pm$ 0.011	0.675
Ours	DMCL-HFM	0.688 $\pm$ 0.011	0.676 $\pm$ 0.009	0.697 $\pm$ 0.014	0.686

5.3. Results of ablation study

To evaluate the effectiveness of the key components in DMCL-HFM, we performed ablation studies by removing specific components. The results of these experiments are shown in Table 3 and Table 4. After removing the Max-Mask strategy, the overall average decreased by 1.04 % and 1.13 % for the cancer subtype classification task and survival prediction task, respectively. This indicates that the Max-Mask strategy can effectively prevent the model from overfitting, thus achieving a better result. Removing the contrast loss led to decreases of 1.15 % and 1.14 % in the overall average for the cancer subtype classification task and the survival prediction task, respectively. This indicates that the contrast loss is crucial for aligning node- and hyperedge-masked augmented hypergraphs, enabling the model to effectively capture both global and local features. Removing the reconstruction loss resulted in a significant decrease in the metrics, which highlights the importance of reconstruction loss in capturing intricate relationships across different levels of the hypergraph. (Table 5)

To verify the effectiveness of the dual-mask strategy, we further compared several variants with different mask strategies, including: 1) Dual-Mask: This variant used both node and hyperedge masking strategies to perform SSL for HGNN pre-training. 2) Node-Mask: This variant only used node masking strategy to perform SSL for HGNN pre-training. 3) Hyperedge-Mask: This variant only used hyperedge masking strategy to perform SSL for HGNN pre-training. Note that a single masking strategy does not support the implementation of contrastive learning. For a fair comparison, the contrastive learning is not incorporated in the Dual-Mask. The results are shown in Fig. 4 and 5. It shows that Dual-Mask performs better than Node-Mask and Hyperedge-Mask on two downstream tasks, indicating it can effectively capture the complex spatial dependencies and contextual interactions among patches. Furthermore, Dual-Mask can provide different views for contrastive learning, thereby further improving the performance of the HFM. In addition, Hyperedge-Mask achieves better results than Node-Mask on most datasets. This indicates that the high-order information contained

Table 4

The classification results of the ablation experiments. The best ones are in bold.

Methods	BRCA		NSCLC		RCC		Overall average
	ACC	AUC	ACC	AUC	ACC	MCC	
W/o Max-Mask	87.55 $\pm$ 2.00	94.26 $\pm$ 1.45	90.63 $\pm$ 1.26	92.48 $\pm$ 1.29	92.89 $\pm$ 1.07	87.90 $\pm$ 2.01	90.95
W/o Contrast. Loss	88.02 $\pm$ 1.51	94.38 $\pm$ 1.34	89.70 $\pm$ 1.69	92.17 $\pm$ 1.47	92.68 $\pm$ 1.09	88.08 $\pm$ 1.49	90.84
W/o Recons. Loss	86.27 $\pm$ 1.20	93.60 $\pm$ 1.24	87.84 $\pm$ 1.29	90.63 $\pm$ 1.47	91.69 $\pm$ 1.30	86.36 $\pm$ 1.12	89.40
DMCL-HFM	88.87 $\pm$ 1.23	95.07 $\pm$ 1.70	91.51 $\pm$ 1.55	93.37 $\pm$ 1.46	93.74 $\pm$ 1.80	89.37 $\pm$ 1.01	91.99

Table 5

The CI values of the ablation experiments. The best ones are in bold.

Methods	IDC	KIRC	KIRP	LUAD	LUSC	Overall average
W/o Max-Mask	0.715 $\pm$ 0.013	0.673 $\pm$ 0.013	0.578 $\pm$ 0.012	0.552 $\pm$ 0.016	0.646 $\pm$ 0.029	0.633
W/o Contrast. Loss	0.708 $\pm$ 0.017	0.675 $\pm$ 0.008	0.581 $\pm$ 0.022	0.559 $\pm$ 0.014	0.640 $\pm$ 0.019	0.632
W/o Recons. Loss	0.689 $\pm$ 0.011	0.657 $\pm$ 0.021	0.567 $\pm$ 0.020	0.550 $\pm$ 0.019	0.627 $\pm$ 0.016	0.620
DMCL-HFM	0.725 $\pm$ 0.018	0.684 $\pm$ 0.013	0.602 $\pm$ 0.013	0.568 $\pm$ 0.011	0.653 $\pm$ 0.014	0.646

in the hyperedge is of greater importance compared to the low-order information in individual nodes. Therefore, the hyperedge can effectively represent the joint relationship among multiple nodes, thereby providing a more comprehensive and accurate representation.

5.4. Hyperparameter analysis

In DMCL-HFM, a fraction of node and hyperedge features in the input hypergraph is randomly masked according to the mask ratio k_1 to generate two augmented hypergraphs for reconstruction and contrastive learning. To investigate the impact of k_1 on model performance, we conducted a series of experiments by gradually increasing k_1 from 0.15 in increments of 0.15. The results are summarized in Fig. 6.

It can be observed that the model performance improves as k_1 increases, reaching a peak around $k_1=0.75$, and then starts to decline. This suggests that moderate feature masking encourages the model to learn more invariant and robust representations. However, when the mask ratio is too high, the augmented views become excessively distorted, making it difficult for the model to align positive pairs effectively, which in turn degrades performance.

5.5. Robustness analysis under different feature extractors

To evaluate the robustness of DMCL-HFM with respect to different patch-level feature extractors, we conducted ablation experiments on the TCGA-RCC dataset, which covers both subtype classification and survival prediction tasks. Specifically, we compared the performance of the model when using features extracted by three representative extractors: ResNet50 (ImageNet pre-trained) [42], UNI (pathology-specific pre-trained) [43], and CTransPath (pathology-specific pre-trained) [31]. As reported in Table 6, DMCL-HFM consistently achieved superior performance regardless of the choice of feature extractor, which confirms the strong generalization capacity of our DMCL-HFM.

We further observed that ResNet50, despite its popularity, exhibited noticeably inferior performance compared to the pathology-specific extractors UNI and CTransPath. This discrepancy highlights the impact of domain-specific pretraining, as ResNet50's natural image priors are less effective for histopathological image representation. In contrast, features from UNI and CTransPath better capture the morphological patterns in WSIs, leading to improved downstream performance. Notably, when integrated with our DMCL-HFM, all feature extractors, regardless of their original training domain, achieved

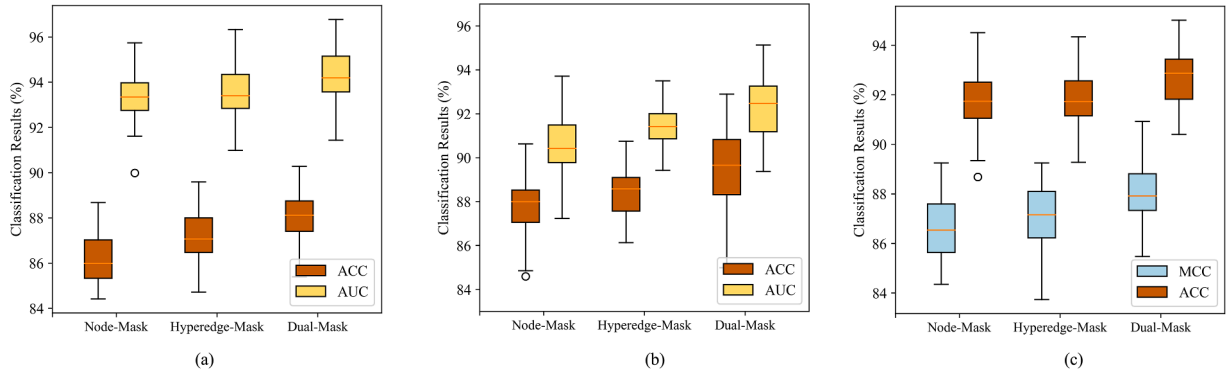


Fig. 4. Classification results under different mask strategies on TCGA-BRCA (a), TCGA-NSCLC (b), and TCGA-RCC(c).

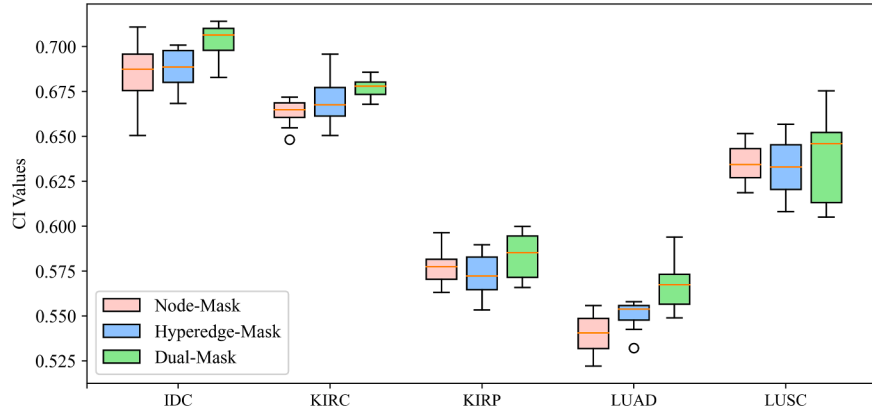


Fig. 5. CI values of different mask strategies on different datasets.

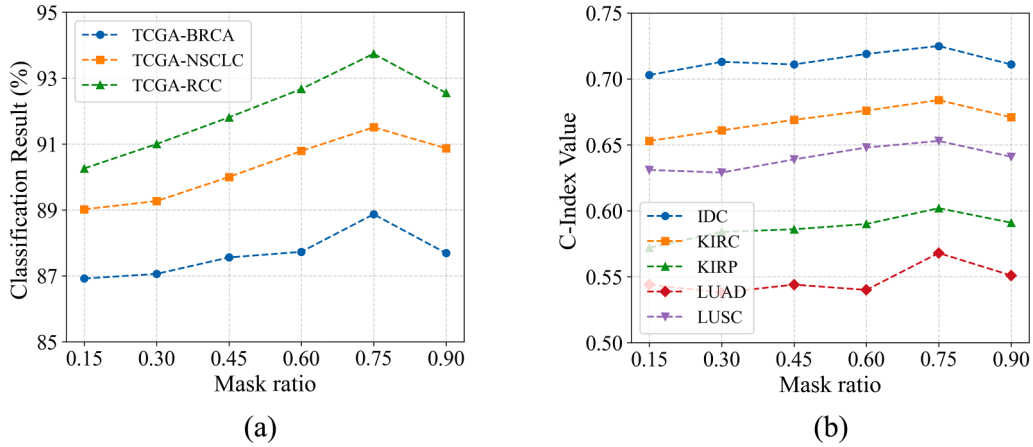


Fig. 6. The results of experiments with varying k_1 values, where (a) shows the performance on the classification task and (b) shows the performance on survival prediction.

substantial performance improvements. The disparities in performance between different extractors were also markedly reduced. This finding highlights the robustness of DMCL-HFM and its adaptability in effectively leveraging diverse feature representations. Such flexibility is particularly advantageous in real-world applications where the choice of feature extractors may be constrained by available computational or data resources.

6. Discussion

In this work, we propose a novel hypergraph foundation model,

DMCL-HFM, which utilizes dual-masked contrastive learning for pre-training on large-scale, unlabeled WSIs. This model can capture both local and global relationships within WSIs, which are essential for understanding tissue structures in WSIs. The extensive experiments demonstrate that DMCL-HFM outperforms existing graph- and hypergraph-based methods, showing its effectiveness in WSI analysis across multiple tissues and diseases.

While most foundation models for WSIs, such as those based on vision transformers, focus primarily on extracting patch embeddings and capturing global features, our method leverages the hypergraph-based structures to capture more complex relationships. This approach

Table 6

Performance comparison of different feature extractors on TCGA-RCC dataset.

Extractors	HGNN Weights	TCGA-RCC		KIRC	KIRP
		ACC	MCC	C-Index	C-Index
ResNet50	Random initialized	85.78 $\pm$ 1.16	75.12 $\pm$ 1.78	0.626 $\pm$ 0.011	0.548 $\pm$ 0.016
		90.07 $\pm$ 1.49	84.22 $\pm$ 1.96	0.633 $\pm$ 0.019	0.550 $\pm$ 0.018
CTransPath		92.42 $\pm$ 1.54	87.02 $\pm$ 1.83	0.656 $\pm$ 0.014	0.562 $\pm$ 0.013
		91.07 $\pm$ 1.64	85.99 $\pm$ 1.91	0.669 $\pm$ 0.012	0.586 $\pm$ 0.015
ResNet50	DMCL-HFM	92.11 $\pm$ 1.59	86.24 $\pm$ 1.66	0.673 $\pm$ 0.017	0.590 $\pm$ 0.017
		93.74 $\pm$ 1.80	89.37 $\pm$ 1.01	0.684 $\pm$ 0.013	0.602 $\pm$ 0.013

captures higher-order dependencies that are crucial for understanding tissue organization in WSIs. Compared to previous hypergraph pre-training methods that mainly use contrastive learning for global feature alignment, DMCL-HFM addresses a key limitation: these methods often miss important local relationships, which are crucial for WSI analysis. DMCL-HFM introduces a dual-masked contrastive learning approach, where node and hyperedge features are masked and reconstructed separately. This enables the model learn both local relationships and global features, making it more suitable for WSI analysis.

However, there are still limitations to the current model. DMCL-HFM is designed for single-modality WSI data, which restricts its application in multi-modal tasks. Future work will focus on extending the model to handle multi-modal data. This will make the model more versatile and applicable to a broader range of tasks in computational pathology.

7. Conclusion

In summary, we propose a novel dual-masked contrastive learning-based hypergraph foundation model for WSIs. We propose a node-hyperedge dual-masked hypergraph modeling strategy, where certain node features are masked and reconstructed using hyperedge features and vice versa. This enables the model to capture intricate local relationships among WSI patch embeddings. By treating the generated hypergraphs as augmentations within a contrastive learning framework, our method effectively captures both local and global features. Additionally, a max-mask strategy reduces overfitting during pre-training, which enhances the model robustness. Extensive experiments on cancer survival prediction and subtype classification show that DMCL-HFM outperforms existing methods, demonstrating its effectiveness in WSI analysis.

CRedit authorship contribution statement

Xueying Zhou: Conceptualization, Methodology, Software, Writing – original draft. **Saisai Ding:** Data curation, Software, Visualization, Writing – original draft. **Wenhua Zhang:** Investigation, Formal analysis, Writing – review & editing. **Juncheng Li:** Writing – review & editing. **Jun Wang:** Writing – review & editing. **Jiasheng Chen:** Writing – review & editing, Data curation. **Jun Shi:** Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All the data we used are publicly available.

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